



S DHEEPAK

B.Tech – Electronics & Communication Engineering

CONTACT

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CORE STRENGTHS

Embedded Systems

Edge AI & Computer Vision

IoT System Design

Industrial Automation

RF & Wireless Systems

Machine Learning

Technical Leadership

Research & Innovation

PROGRAMMING

Embedded C

C++

Python

HTML5 / CSS3

JavaScript

MATLAB

PROFILE

Passionate Electronics and Communication Engineering undergraduate proficient in Embedded Systems, STM32, ESP32, Edge AI, Machine Learning, Computer Vision, IoT System Design, RF Systems, and Industrial Automation. Experienced in developing end-to-end AI-driven, real-time embedded solutions for intelligent transportation, disaster management, and industrial systems. Proven track record of innovation with an Indian Patent publication, IEEE research publication, and active involvement in technical leadership and student organizations.

EDUCATION

Kalasalingam Academy of Research and Education

B.Tech – Electronics and Communication Engineering

CGPA: 9.03 | 2023–2027 | Expected Graduation:

May 2027

Velammal Vidyalaya, Paruthipattu, Avadi, Chennai

Senior School Certificate Examination (CBSE) Class XII – Computer Science stream

2023 | Result: PASS

Velammal Vidyalaya, Paruthipattu, Avadi, Chennai

Secondary School Examination (CBSE)

2021 | Result: PASS

TECHNICAL SKILLS

- **Embedded:** STM32, ESP32, ESP32-S3, ATmega328P, CAN Protocol, HIL Validation Concepts, Sensor Fusion
- **AI & Computer Vision:** Edge AI, DeepFace, OpenCV, Machine Learning, PSO, Feature Extraction
- **IoT & Wireless:** LoRa RA-02, LoRa Mesh, Fog Computing, ThingSpeak Cloud, Sensor Networks
- **Languages & Tools:** Embedded C, C++, Python, HTML5, CSS3, JavaScript, Tkinter, MATLAB, Proteus, RF Smith Chart

PATENTS

PyroSentinel – AI-Powered LoRa-Based Low Power Forest Fire Early Detection and Intelligent Communication System Using PSO-Optimized Mesh Architecture

Application No.: 202641055047 A | Publication: June 26, 2026

TOOLS / PLATFORMS

OpenCV / DeepFace

STM32

ESP32 / ESP32-S3

ThingSpeak

Cloud

MATLAB

Proteus

Tkinter

RF Smith Chart

LEADERSHIP

Public Relations Team Lead – ACM
SIGBED Student Chapter (2025–
2026)

Student Coordinator – Embedded
Systems Industry Session, NanoChip
Solutions

Student Coordinator – Guest
Lecture on HDL in VLSI, Elevium

Student Coordinator – Euphoria /
IoTRONZ Challenge & VINTRAA
Technical Event

Public Manager – DIGICON 4.0

NSS Volunteer

Offline LoRa–Fog Intelligent Safety Watch Network with Particle Swarm Optimization for Automated Emergency Interception (ABHAYA)

Patent Filed: July 2026 | Offline wearable safety and
rescue communication system

Interactive Web–Based Smith Chart Analyzer with Integrated Offline LLM Assistance for Real–Time RF Impedance Matching and Visualization

Patent Filed: May 2026 | Interactive RF impedance
matching platform

PUBLICATION

An Adaptive Radiomics–Guided Spatial Clustering Framework for Lung Tumor Segmentation from CT Images Publisher: IEEE (IEEE Xplore) | Authors: M. Ganga Priyadarshini, Dr. V. Muneeswaran, S. Dheepak

- Proposed an adaptive radiomics–guided spatial clustering framework for automated lung tumor segmentation from CT scans.
- Applied edge–preserving filtering and CLAHE followed by morphological refinement.
- Integrated entropy–based spatial clustering and Adaptive Fuzzy C–Means (FCM) without requiring prior training data.

SELECTED PROJECTS

Edge AI–Based Driver Emotion Recognition and Adaptive Vehicle Control System

**Technologies: Python, DeepFace, OpenCV, ESP32,
Embedded C, CAN, HTML, CSS, JavaScript, Edge AI**

- Built a real–time pipeline to classify driver states such as Calm, Aggressive, and Fatigued.
- Interfaced ESP32 control algorithms to adjust vehicle speed and initiate emergency stops on fatigue detection.
- Developed an Edge AI–powered live web dashboard for real–time driver–state monitoring.

Autonomous Robotic Arm for Material Handling using STM32

Technologies: STM32, Embedded C, Motor Drivers, Sensors, Industrial Automation

- Designed an autonomous robotic arm for transporting materials between industrial workstations.
- Implemented precise motion-control and reliable pick-and-place routines using Embedded C.

IoT-Based Biometric Attendance Monitoring System

Technologies: ESP32-S3, R307S Fingerprint Sensor, OLED, C++ / Embedded C, IoT

- Developed fingerprint-based authentication with real-time OLED feedback.
- Engineered a compact hardware enclosure and cloud-ready architecture for remote data logging.

TECHNICAL PROJECTS

Wearable Non-Invasive Glucose Monitoring System – Nanochip Hackathon

Technologies: Python, Tkinter, ThingSpeak Cloud, Excel Analytics, Data Visualization

- Engineered a wearable-oriented non-invasive glucose monitoring prototype during a 24-hour continuous hackathon sprint.
- Processed and analyzed a simulated dataset of over 2,000 glucose data points for trends and statistical insights.
- Integrated ThingSpeak cloud architecture for remote tracking and developed a desktop GUI for live feedback.

Interactive Web-Based Smith Chart Analyzer

Technologies: HTML, CSS, JavaScript, RF Engineering

- Developed an interactive application for RF impedance matching and high-frequency circuit analysis.
- Implemented automated VSWR and complex impedance matching calculations.
- Integrated offline LLM-assisted RF analysis features for microwave network interpretation.

EXPERIENCE & LEADERSHIP

Public Relations Team Lead — ACM SIGBED Student Chapter (2025–2026)

Spearheaded PR and outreach for South India's first ACM SIGBED Student Chapter, gaining coverage in regional and national outlets. Managed chapter showcase stalls highlighting student engineering innovations to industry experts.

Student Coordinator — Embedded Systems Industry Session, NanoChip Solutions (March 2025)

Coordinated the session "Embedded System: The Power of Bits in Action and Validation Approaches for Automotive ECU," featuring Mr. S. Vignesh, HIL Validation Engineer.

Student Coordinator — Guest Lecture on HDL in VLSI, Elevium (October 2025)

Organized a guest lecture by Mr. Niranjana Sharma, Design & Verification Engineer, covering SystemVerilog, UVM methodologies, and SoC protocols including AXI, AHB, and PCIe.

Student Coordinator — Euphoria / IoTRONZ Challenge & VINTRAA Technical Event (2025)

Managed event logistics, planning, and student participation across multiple campus technical challenges.

Public Manager — DIGICON 4.0

Directed public communication strategies and organized awareness campaigns on emerging technologies.

NSS Volunteer — National Service Scheme

Participated actively in community welfare activities, blood donation camps, and social service initiatives.

AWARDS & HONORS

- IEEE Published Author – Radiomics-Guided Spatial Clustering for Lung Tumor Segmentation (2026)
- Published Indian Patent Holder – PyroSentinel AIoT Forest Fire Early Detection System (2026)
- AICTE IIC Regional Meet 2025 – Shortlisted for YUKTI & APF Grant Competitions (Round 2)
- IASF EduAITHon 2026 – GITAM University
- ELECTROTHON 2026 – Rathinam Technical Campus
- Ideathon 2026 – PSR Engineering College
- Nanochip Hackathon 2025 – Nanochip Industries Pvt. Ltd.

SELECTED ACHIEVEMENT HIGHLIGHTS

- IEEE research publication in lung tumor segmentation using adaptive radiomics-guided spatial clustering.
- Indian patent publication for PyroSentinel, a PSO-optimized LoRa mesh forest-fire detection and communication system.
- Shortlisted at AICTE IIC Regional Meet 2025 for YUKTI & APF Grant Competitions (Round 2).
- Active participation in IASF EduAIThon 2026, ELECTROTHON 2026, Ideathon 2026, and Nanochip Hackathon 2025.